QUANT + TRADING

WEEK 05 VAULT

Five open-source quant tools the funds actually use — each with a ready-to-paste prompt to start backtesting today.

JUNE 2026 · 5 PICKS · EN / AR

> LEAN > NautilusTrader > vectorbt > Freqtrade > OpenBB

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Welcome to the Vault.

This week it's quant: the open-source engines, backtesters, and data layers that serious trading desks run on — not the paid black-box bots filling your feed. Each pick is a real repo you clone and learn from, with what it does, why it matters, the GitHub link, and a prompt you paste straight into your AI. These are tools to research and backtest ideas with — for learning, not financial advice, and nobody here is promising profit.

▸ **HOW TO USE EACH PICK**

1. Read the hook — decide in 5 seconds if it's for you.
2. Open the GitHub link.
3. Copy the **>_ DROP THIS INTO YOUR AI** prompt into Claude, ChatGPT, or Codex.
4. Let it walk you through setup & first use. No guesswork.

هالأسبوع كوانت: المحركات والباك-تيست والبيانات مفتوحة المصدر اللي تشتغل عليها مكاتب التداول الجادة — مش البوتات المدفوعة الصندوق-الأسود اللي تشوفها بكل مكان. كل أداة ريبو حقيقي تكلونه وتتعلم منه: شرح، وليش تهمك، ورابط GitHub، وبرومبت تنسخه في Claude أو ChatGPT أو Codex. هذي أدوات تبحث وتختبر فيها أفكارك — للتعلّم، مش نصيحة مالية، وما أحد يوعدك بربح.

THIS WEEK

01	LEAN	TRADING ENGINE · FUND-GRADE
02	NautilusTrader	RUST-NATIVE · EVENT-DRIVEN
03	vectorbt	VECTORIZED BACKTEST · FAST
04	Freqtrade	STRATEGY AUTOMATION · BOT
05	OpenBB	DATA LAYER · 350+ SOURCES

QUANT / ENGINE · QUANTCONNECT

LEAN

01

★ ~12k · powers QuantConnect · 350k+ quants · verdict RUN

The same open-source engine quant funds run — free. Backtest and live-trade across stocks, forex, crypto, and futures from one codebase in C# or Python, all on your own machine.

نفس المحرك مفتوح المصدر اللي تشغّله صناديق الكوانت — مجاني. باك-تيست وتداول مباشر على الأسهم والفوركس والكريبتو والعقود من كود واحد بـ #C أو Python، كله على جهازك.

WHAT IT IS

LEAN is QuantConnect's algorithmic trading engine — the core that runs their cloud platform, open-sourced for anyone to run locally. You write one strategy class, and the same code backtests on years of historical data and then trades live, with built-in handling for portfolio construction, risk, order routing, and multiple asset classes. You drive it with the Lean CLI: `pip install lean`, and it spins up a Docker container with the whole engine and data stack inside, so there's nothing to compile by hand.

WHY IT MATTERS FOR YOU

Most retail tooling is a toy compared to this — LEAN is the kind of event-driven, multi-asset architecture real desks depend on, and you get it for the cost of cloning a repo. Learn how a serious engine separates strategy from execution and you understand how the pros actually build. (It's a research and backtesting tool — not a money printer.)

repo > github.com/QuantConnect/Lean

>_ DROP THIS INTO YOUR AI

```
Set up QuantConnect LEAN (github.com/QuantConnect/Lean) locally on Windows so I can
backtest a strategy. Give me exact steps: install Docker, `pip install lean`, run `lean
init`, create a Python project, and run a backtest on sample data. Then explain how the
same algorithm class moves from backtest to live, and flag any prerequisite I'm missing.
```

QUANT / ENGINE · NAUTECH

NautilusTrader

02

★ ~17k · Rust core · Python API · verdict RUN

A trading engine with a Rust core and a Python face: nanosecond-accurate event handling, so the strategy you backtest is the exact same code you run live — no rewrite, no drift.

> محرك تداول قلبه Rust وواجهته Python: معالجة أحداث بدقة نانوثانية، فالاستراتيجية اللي تختبرها هي نفس الكود اللي تشغّله مباشر — بدون إعادة كتابة وبدون انحراف.

WHAT IT IS

NautilusTrader is a production-grade engine built in Rust for speed and safety, with a Python layer where you actually write strategies. Its big idea is parity: the event-driven backtester replays history tick by tick the same way the live system processes a real feed, so what you test is genuinely what you deploy. It's multi-venue and multi-asset, with a deterministic core that makes results reproducible — the same inputs give the same outputs every run.

WHY IT MATTERS FOR YOU

The number one way backtests lie is when live code differs from test code; Nautilus is engineered specifically to close that gap. The Rust core also means you can handle high-frequency data without Python choking. It's a sharper, lower-level complement to LEAN — pick the one whose architecture clicks for you. (Backtest parity is about honesty, not guaranteed returns.)

repo > github.com/nautechsystems/nautilus_trader

>_ DROP THIS INTO YOUR AI

```
Help me get started with NautilusTrader (github.com/nautechsystems/nautilus_trader) for
event-driven backtesting in Python. Walk me through installing it, loading some
historical bar data, writing a minimal strategy class, and running a backtest. Then
explain how its backtest-to-live parity works and what 'event-driven' actually changes
versus a simple loop.
```

QUANT / RESEARCH · PYTHON

vectorbt

03

★ ~5k · NumPy + Numba · thousands of tests/sec · verdict BOTH

Test thousands of strategy variations in seconds, not hours. vectorbt packs every parameter combo into NumPy arrays and runs them all at once — grid search that used to take all afternoon finishes before your coffee's cold.

> اختبر آلاف نسخ الاستراتيجية بثواني، مش ساعات. vectorbt يحزم كل توليفات المعطيات في مصفوفات NumPy ويشغّلها دفعة وحدة — بحث الشبكة اللي كان ياخذ عصر كامل يخلص قبل ما يبرد قهوتك.

WHAT IT IS

vectorbt is a Python research library that takes a radically different angle on backtesting: instead of looping bar by bar through one strategy at a time, it vectorizes everything into NumPy arrays and accelerates the hot path with Numba. That lets you sweep huge parameter grids — hundreds of moving-average crossovers, dozens of assets, multiple timeframes — in a single fast pass, then drill into trades, drawdowns, and stats, with QuantStats tearsheets built in.

WHY IT MATTERS FOR YOU

When you're hunting for an edge, speed of iteration is everything — the more ideas you can test honestly, the faster you learn what doesn't work. vectorbt is the research bench that sits in front of an execution engine like LEAN or Nautilus: explore here, validate the survivors there. The open-source community edition is plenty to learn the whole workflow. (Fast backtesting finds patterns; it doesn't promise they'll hold.)

repo > github.com/polakowo/vectorbt

>_ DROP THIS INTO YOUR AI

Teach me vectorbt (github.com/polakowo/vectorbt) by example. Install it, pull some free daily price data, and build a moving-average crossover backtest. Then show me the killer feature: sweep a grid of fast/slow window combos in one vectorized run and rank them by return and drawdown. Explain why vectorizing is so much faster than a per-bar loop.

QUANT / AUTOMATION · PYTHON

Freqtrade

04

★ ~40k · 30+ exchanges · Telegram control · verdict BOTH

The most battle-tested open-source trading bot there is. Write a strategy in Python, backtest it, dry-run it on live prices with fake money, and control the whole thing from Telegram — all without touching a paid platform.

> أكثر بوت تداول مفتوح المصدر مُجرّب فعليًا. اكتب استراتيجية بـ Python، اختبرها، شغّلها dry-run على أسعار حقيقية بفلوس وهمية، وتحكّم بكل شي من Telegram — بدون أي منصة مدفوعة.

WHAT IT IS

Freqtrade is a free, open-source crypto trading bot that bundles the full lifecycle: backtesting, hyperparameter optimization, a web dashboard, and Telegram control in one tool. Its safest feature is dry-run mode — the bot runs against real, live market prices but trades with simulated money, so you can watch a strategy behave in real conditions before risking anything. It supports 30+ exchanges and a huge library of community strategies to read and learn from.

WHY IT MATTERS FOR YOU

This is where research meets reality: a strategy that looks great in a backtest still has to survive live spreads, fees, and slippage, and dry-run mode lets you witness that gap safely. Even if you never go live, Freqtrade is one of the best ways to learn how an automated strategy is structured end to end. Start in dry-run, stay in dry-run as long as you're learning. (This is education and tooling — not a signal service, and not financial advice.)

repo > github.com/freqtrade/freqtrade

>_ DROP THIS INTO YOUR AI

```
Set up Freqtrade (github.com/freqtrade/freqtrade) in safe dry-run mode only — I'm learning, not risking real money. Walk me through installing it via Docker, generating a config, downloading historical data, backtesting a sample strategy, and then launching dry-run mode against live prices with simulated funds. Show me how to monitor it from Telegram and read the results.
```

QUANT / DATA · OPENBB

OpenBB

05

★ ~40k · equities · crypto · macro · SEC filings · verdict BOTH

Every engine above needs clean data — OpenBB is the open-source pipe that delivers it. One Python interface to 350+ financial datasets: equities, forex, crypto, economics, and SEC filings, no scattered API keys to juggle.

> كل محرك فوق يحتاج بيانات نظيفة — OpenBB هو الأنبوب مفتوح المصدر اللي يجيبها. واجهة Python وحدة لأكثر من ٣٥٠ مصدر بيانات: أسهم، فوركس، كريبتو، اقتصاد، وملفات SEC — بدون ما تطارد مفاتيح API متفرقة.

WHAT IT IS

OpenBB is an open-source financial data platform for analysts, quants, and AI agents. Instead of writing a fresh wrapper for every vendor, you call one consistent Python SDK and it normalizes data from hundreds of providers — equities, options, crypto, macroeconomic series, alternative data, and regulatory filings — into clean, ready-to-use objects. It plugs straight into pandas, so the output drops directly into a vectorbt or LEAN workflow.

WHY IT MATTERS FOR YOU

Garbage in, garbage out — the most overlooked part of any quant stack is reliable, normalized data, and that's exactly the unglamorous job OpenBB does well. It's the foundation the other four picks sit on: pull and clean here, research and backtest there. Free, AI-agent friendly, and it spares you the worst part of every project — wiring up data sources by hand. (Good data sharpens analysis; it doesn't guarantee the trade.)

```
repo > github.com/OpenBB-finance/OpenBB
```

>_ DROP THIS INTO YOUR AI

```
Get me started with the OpenBB Platform (github.com/OpenBB-finance/OpenBB) as a data
layer for quant research. Install the Python SDK, then show me how to pull historical
equity prices, a macro series, and crypto data into pandas DataFrames with a consistent
interface. Explain how to add a free data-provider API key, and how this output feeds
cleanly into a backtesting library like vectorbt.
```

That's the Vault for this week.

Five tools that form one stack: an engine, two ways to research and backtest, an automation layer, and the data underneath it all. Pick one, clone it, run a backtest this week — that's the whole point.

Next week: more under-surfaced quant & data picks — alpha research frameworks, market-data pipelines, and the risk layer most retail traders skip.

Found one of these useful? Forward this to one person who'd get it. That's how the Vault grows.

خمس أدوات تركّب ستاك واحد: محرك، طريقتين للبحث والباك-تيست، طبقة أتمتة، والبيانات تحت كل هذا. اختر وحدة، كلونها، شغّل باك-تيست هالأسبوع — هذا كل الهدف. عجبتك أداة؟ ابعثها لشخص يستفيد. هكذا تكبر الخزينة.